Isman Kurniawan
School of Computing, Telkom University
Bandung, Indonesia
isman.krn@gmail.com

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To the Editors, *Journal of Chemical Information and Modeling*

Dear Editors,

We submit for your consideration an Article entitled **"Activity cliffs on human acetylcholinesterase: a pre-specified descriptor escalation"**.

It is widely held that two-dimensional fingerprints fail on activity cliffs because they cannot see the three-dimensional detail separating the two members of a pair, and that supplying that detail should help. We tested that directly, escalating four structure-based representations against the same fingerprint baseline on the same frozen splits, and measured in advance how large a difference the design could have resolved. None was resolvably better on any of three framings of the endpoint, and on per-ligand potency the molecular dynamics descriptors were resolvably worse. The result is bounded rather than open, which is what separates it from an underpowered null.

Three findings carry beyond this target. A replicate test separates persistence from amplitude sharply, so a trajectory of affordable length reproduces how long a contact is held and not how far things move. A descriptor filter refitted at each value of a varied parameter manufactures a trend that a single fixed selection removes, which exposes any study that re-selects features as it varies a parameter. And the descriptors that survive reproducibility are available from 15 ns of simulation rather than the 50 the original design called for.

We would like to address three points of the Journal's guidance on QSAR submissions before a reader reaches them. First, this is not a docking study against a single target: the docking serves a comparison between representations, and the quantity under test is what physical detail adds to a matched ligand-based baseline. Second, the high ratio of descriptors to data points sits in the fingerprint baseline, scored as a practitioner would score it, and we report a matched-dimensionality arm alongside it precisely so that feature count can be separated from representation. Third, we are not proposing new descriptors and claim no superiority for them; what is offered is the comparison and the detection limit that makes its outcome readable, and the descriptor machinery is standard and attributed as such.

Acetylcholinesterase was chosen because the question needs a target supplying activity cliffs in quantity and a crystallographic human receptor, and because its gorge gates conformationally on evidence predating this work. A comparison that finds no advantage where one is most expected says more than one that finds none where none was expected.

All molecular structures appear in the Supporting Information, and all analysis code and derived data are in a public repository archived with a DOI, sufficient to reproduce every number without rerunning a simulation.

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors. The authors declare no competing financial interest, and the work received no external funding.

Yours sincerely,

Isman Kurniawan, on behalf of all authors